Luca Ziviani

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SUMMARY:

Applied Mathematics PhD candidate specializing in PDEs, numerical simulation, and high-dimensional modelling. Experienced in Python/C++ scientific computing, optimization, and large-scale numerical experiments on HPC clusters. Interested in quantitative research, AI, scientific computing, and mathematically intensive engineering roles.

TECHNICAL SKILLS

Programming:

- Python (NumPy, SciPy, PyTorch, OpenCV);
- C++;
- MATLAB;
- LaTeX.

Scientific Computing:

- Numerical PDEs;
- Optimization;
- Monte Carlo;
- High-dimensional modelling.

Machine Learning:

- PyTorch;
- Optimization methods;
- Data analysis;
- HPC clusters.

Mathematics:

- Probability;
- Statistics;
- Linear Algebra;
- Numerical Analysis.

EXPERIENCE

2022 - 2026; **Reseacher in Applied Mathematics** | PhD candidate
Supervisor: Emeric Bouin *Laboratory*: CEREMADE, Université Paris-Dauphine PSL.

2022 - 2026; **Teaching Assistant.** *Establishment*: MIDO, Université Paris-Dauphine PSL.
- Analysis 1, Bachelor's Degree in Applies Mathematics, 1st year.
- Analysis 2, Bachelor's Degree in Applies Mathematics, 1st year.
- Game Theory, Bachelor's Degree in Applies Mathematics, 3rd year.

PROJECTS

- Numerical simulation of the *kinetic Fokker-Planck equation* for gas and plasma. Link to GitHub. Python.
- Numerical simulation of the *run and tumble equation* for chemotaxis. Link to GitHub. Python.
- Jaguar: Energy-efficient navigation of a terrestrial robot in an uncertain environment. Link to GitHub. Python (JAX).
- Web application to play my collection of crosswords and puzzles. Link to the page. HTML, CSS, JavaScript.

SCIENTIFIC PUBLICATIONS

[1] *L^2 Hypocoercivity methods for kinetic Fokker-Planck equations with factorised Gibbs states*.

[2] *Sub-exponential tails in biased run and tumble equations with unbounded velocities*.

[3] *Convergence to a non-explicit steady state in non-factorized kinetic Fokker-Planck equations with (very) weak velocity confinements*.

- *Developed analytical proofs and implemented large-scale numerical PDE simulations on HPC infrastructure.*

EDUCATION

2022 - 2026; **PhD, Mathematics**, Université Paris-Dauphine PSL (Paris).
Supervisor: Emeric Bouin *Laboratory*: CEREMADE

2020 - 2022; **Master degree, Mathematics**, MAPPA double degree program operated by University of Padova (Italy) and University Paris Sciences & Lettres (France)
Title: Kinetic Theory and Hypocoercivity. *Final grade*: 110/110 cum laude.

2017 - 2020; **Bachelor degree, Mathematics**, Università di Padova, Italy.
Title: Rademacher and Whitney's Theorems *Final grade*: 110/110 cum laude

2013 - 2017; **High school degree**, Liceo scientifico Copernico-Pasoli, Verona, Italy.

LANGUAGES

English (fluent), French (fluent), Italian (native)

INTERESTS

Piano, Athletics, Enigmatics.